Amar Pratap Singh

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Education

Kalinga Institute of Industrial Technology (KIIT) Bhubaneswar, Odisha
Bachelor of Technology in Computer Science and Engineering; CGPA: 7.97/10 *2023 – Present*

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Cloud Computing

Research Experience

Explainable Artificial Intelligence for Transparent Medical Diagnosis
Research Co-author & Presenter — ComSIA-2026 *Mar. 2026*

- Conducted literature analysis of XAI techniques in healthcare, including SHAP, LIME, and Grad-CAM, and their application to medical decision-making
- Analyzed research on AI transparency, clinical trust, bias, interpretability, and challenges in applying XAI to healthcare
- Co-authored and presented the paper at ComSIA-2026

Projects

BitGit | *Node.js, SHA-1, Data Structures, Unit Testing* GitHub

- Implemented a Git-like distributed version control system from scratch, designing a content-addressable object database using SHA-1 hashing and two-character directory fan-out for efficient object storage and lookup
- Implemented core version-control operations including init, add, commit, status, diff, checkout, and log, with automated unit tests

AirSocket | *Java 21, TCP/UDP, AES-256-GCM, JUnit 5, Maven* GitHub

- Engineered a P2P file-transfer and network profiling system in Java, diagnosing severe latency overhead from Nagle's algorithm and TCP Delayed ACKs during LAN throughput testing
- Improved local network throughput 5.41×, from 163.6 Mbps to 886.0 Mbps, through socket buffer tuning and application-level chunk-size optimization
- Achieved 89.3 MB/s (714.4 Mbps) end-to-end file-transfer throughput over physical network interfaces while verifying byte-identical file integrity through custom stream chunk verification

ThreadTrace | *Java, Multithreading, ExecutorService, File I/O* GitHub

- Built a concurrent multi-threaded CLI tool in Java that divides large log files into equal byte-sized chunks and processes segments in parallel using ExecutorService and low-level RandomAccessFile pointers
- Designed an isolated, thread-confined state architecture where worker threads independently collect structural metrics (errors and warnings grouped by component, time, and message) before safely merging results into a unified report

Technical Skills

Languages: C, C++, Java, JavaScript
Frameworks/Runtime: Node.js, React.js
Web: HTML, Tailwind CSS
Databases: PostgreSQL, SQL
Systems & Tools: Linux, Git, Docker

Certifications

- **Java Development Certification** — Udemy | Jan 2025: Core Java, OOP, Multithreading, File I/O, Collections